A.2. Calculus

Calculus is essentially the study of differentiation and integration. In deep learning, we only really need to use differentiation. *Differentiation* is the process of getting a derivative of a function.

We already introduced one notion of derivative: the ratio of an output interval to the input interval. It tells you how much the output space is stretched or squished. Importantly, these intervals are oriented intervals so they can be negative or positive, and thus the ratio can be negative or positive.

For example, consider the function $f(x) = x^2$. Take a point x and its neighborhood $(x – \varepsilon, x + \varepsilon)$, where ε is some arbitrarily small value, and we get an interval around x. To be concrete, let $x = 3$, $\varepsilon = 0.1$; the interval around $x = 3$ is (2.9,3.1). The size (and orientation) of this interval is 3.1 – 2.9 = +0.2, and this interval gets mapped to $f(2.9) = 8.41$ and $f(3.1) = 9.61$. This output interval is (8.41,9.61) and its size is 9.61 – 8.41 = 1.2. As you can see, the output interval is still positive, so the ratio $\frac{df}{dx} = \frac{1.2}{0.2} = 6$, which is the derivative of the function f at $x = 3$.

We denote the derivative of a function, f, with respect to an input variable, x, as df/dx, but this is not to be thought of as a literal fraction; it’s just a notation. We don’t need to take an interval on both sides of the point; an interval on one side will do as long as it’s small, i.e., we can define an interval as $(x, x + \varepsilon)$ and the size of the interval is just ε, whereas the size of the output interval is $f(x + \varepsilon) – f(x)$.

Using concrete values like we did only yields approximations in general; to get absolutes we’d need to use infinitely small intervals. We can do this symbolically by imagining that ε is an infinitely small number such that it is bigger than 0 but smaller than any other number in our number system. Now differentiation becomes an algebra problem.